Task 2:

We implemented a semantic analysis check for the variables in Odenc, to detetct errors before program execution. We implemented two checks cvd_program() and rbw_program()

The cvd_program() method checks that every variable used in an expression has been declared, it collects the declared variable name in a set and goes through the statements in the program to check if it fits, if not it raises an undeclared variable error SemCheckUndeclaredVariable.

Likewise for the rbw_program() but this one checks if the variables are read before they have been assigned a value, it keeps a set of initialized variables, and goes through the statements in program order ie For an assignment, it checks the right hand-side expression before marking the target variable as initialized. For a print statement, it checks the expression to ensure that every variable it reads has already been initialized.
If a variable is read too early it raises an error SemCheckUninitializedVariable.n

We tested our implementation first by running ./check.sh like we did in task 1, but it only runs the existing compiler and interpreter tests and does not include SemCheck, so we manually checked it by parsing small programs with LangVar().parse() and calling them directly.

Var x; x = 1; print(x);

passed rbw_program, meanwhile:

var x; print(x); and var x; x = x +1; both raised a SemCheckUnintializedVariable: x

and likewise for var x; x = y +1; made cvd_program() raise SemCheckUndeclaredVariable: y